

**Sixth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

COMPUTER ARCHITECTURE

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Assume suitable data wherever necessary.
- (2) All questions are compulsory.
- (3) Neat sketches and block diagrams are expected wherever necessary.

1. (a) Implement the given functions using PAL and PLA.

$$f_1(a, b, c, d) = \prod M(2, 3, 8, 12, 14, 15).d(10, 11)$$

$$f_2(a, b, c, d) = \sum m(5, 7, 8, 9, 10, 11, 12, 14) \quad 10(\text{CO4})$$

2. Perform Booth Multiplication algorithm on $(-42)_{10}$ and $(-76)_{10}$ along with the flowchart. 10(CO2)

3. Design Control unit for Robertson multiplier in a Computer system using Delay Element Method and Sequence Counter Method. 10(CO5)

4. (a) Describe Priority Control Function in a Bus Control Unit. 5(CO1)

- (b) Perform division on $(15)_{10}$ by $(2)_{10}$ using non-restoring algorithm. 5(CO2)

5. (a) Consider a paging system in which main memory has the capacity of three pages. The execution of a program Q requires reference to find distinct pages P_i where $i = 1, 2, 3, 4, 5$ and i is the page address. The page stream formed by executing Q is

2 3 1 1 5 1 4 4 3 2 5 2

Assess how pages are assigned to main memory using FIFO, LRU and OPT replacement policies. 5(CO3)

- (b) Discuss taxonomy of parallel processor architecture. 5(CO1)
6. (a) Analyze internal organization of CPU with cache memory. 5(CO1)
- (b) Discuss vector arithmetic unit is a computer system with diagram. 5(CO1)

